Chanhyun Park

Email: chanhyeonpark00@gmail.com
Website: https://chanhyunpark-eng.github.io/
Repo: https://github.com/chanhyunpark-eng

Interests

How should floating structures and marine vehicles be designed and controlled so that they stay safe and productive in real sea conditions, where the hull, the waves and the control system cannot be treated separately?

I am interested in: Floating offshore structures, Floating body dynamics, Ocean renewable energy, Marine vehicle dynamics and control, Subsea robotics (ROV), Marine autonomy

Education

2021 – 2027 **Korea Maritime and Ocean University**, (Busan, Korea)
B.S. in Naval Architecture and Ocean Systems Engineering (expected Feb 2027)

- GPA since returning from military service (Spring 2025 – Spring 2026): **4.34** / **4.5**; cumulative GPA 3.73 / 4.5
- Ranked in the top four of the department in each of the last three semesters, including *1st* in Spring 2025 with a 4.5 / 4.5 term GPA
- Relevant coursework: Floating Body Dynamics, Marine Systems Control, Computational Fluid Dynamics, Numerical Analysis, Ocean Structure Design

Professional Experience

09/2026 – 12/2026 **Underwater Solution Co., Ltd.** (Korea)
Intern, R&D Department

- Piloted a Blue Robotics ROV for underwater hull inspections of berthed commercial vessels in port
- Maintained ROVs, fabricated components by 3D printing, and wrote the technical report after each inspection

Projects

2026 **Rim-Driven Tidal Current Generator on a SPAR Platform**
Team leader (team of 4), 2026 Ocean Engineering CAE Competition (national) · Technical report

- Selected the NACA 0018 blade section from QBlade double-multiple-streamtube sweeps over four sections, solidity 0.10–0.50 and TSR 0–8 ⇒ accepted 1.8% lower Cp for a 50% thicker section that the structure required
- Derived the 100-year-current design load (179 kN/m) and sized the GFRP blade shell to DNV-OS-C501; formulated a minimum-mass optimization of the S690 support arms (444 MPa vs 500 MPa allowable)
- Sized the rotor for neutral buoyancy ⇒ net output of 1.09 MW, 9.4% of the platform's 11.6 MW demand, with no added deadweight
- Received the President's Award of The Korean Society of Ocean Engineers

2025 **Payload Limit of a Model-Scale Semi-Submersible**
Co-lead (team of 5), Ocean Structure Design course · Technical report

- Designed a six-column PVC model and set the deck proportions to balance roll and pitch stability
- Traced a tank-test capsize (2–3 payload units vs 6 predicted) to a displaced-volume error that overstated GM about threefold ⇒ corrected analysis predicts loss of stability at 2.9 units, consistent with the test

2025 **Free-Surface CFD of Boat Resistance on a 300,000-Cell Budget**
Individual term project, STAR-CCM+ · Technical report

- Designed a RANS/VOF mesh under a fixed cell budget with no reference solution, checking each decision against an independent physical estimate (wave height, boundary-layer thickness, first-cell y+)
- Attributed a 22% resistance increase from a bow plate to pressure drag ⇒ a friction-line estimate explained only 6% of the increase

Honors & Awards

11/2026 **KMOU Alumni Association President's Award** – one of three students university-wide, representing the College of Ocean Science and Engineering

10/2026 **President's Award, The Korean Society of Ocean Engineers** – 2026 Ocean Engineering CAE Competition (national)

11/2022 **President's Award, The Society of Naval Architects of Korea** – 31st National Student Ship Design Contest (national); hull lines design of a 319,000 DWT VLCC

2025 – 2026 **Merit-based Scholarship**, Korea Maritime and Ocean University – Spring 2025, Fall 2025, Spring 2026

Leadership & Service

03/2026 – current **Student Council President**, College of Ocean Science and Engineering, Korea Maritime and Ocean University

06/2023 – 12/2024 **Republic of Korea Army** – Drill Instructor, Korea Army Training Center

Skills

Analysis	STAR-CCM+ (RANS, VOF), QBlade
Design	Rhino, HForm, AutoCAD
Programming	Python (NumPy, SciPy, Matplotlib), Arduino
Languages	Korean (native), English (TOEFL iBT 90)